



Applied Data Mining

**— Week 7—
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Concept Hierarchy in Data Mining

Organizing Data for Better
Analysis and Pattern
Discovery

Introduction & Motivation



If you have 1 million rows of data, can you analyze it directly?

No → too complex

- Data is **too detailed** → hard to analyze
- We need **simplification without losing meaning**

Organizing Data for Better Analysis and
Pattern Discovery

What is a Concept Hierarchy?

- Organizes data into multiple levels of abstraction

- From low-level detailed values to high-level concepts

- Supports drill-down (detailed view) and roll-up (summary view)

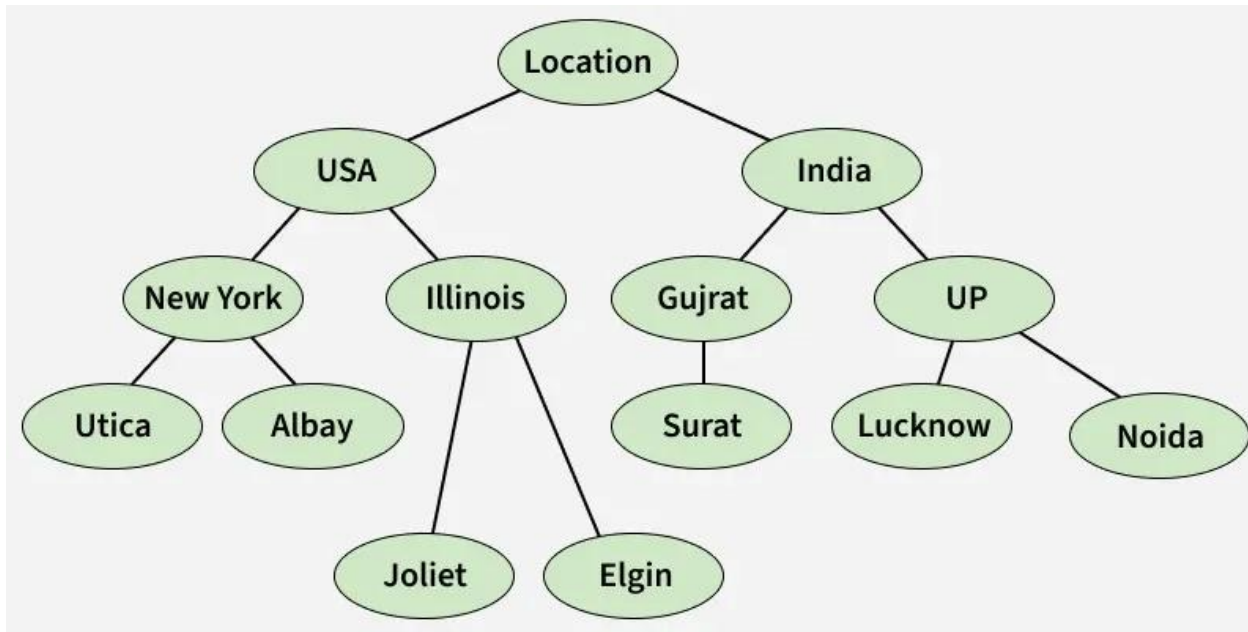
- Simplifies large datasets and improves pattern discovery

- A concept hierarchy organizes data into multiple levels of abstraction-
- ranging from detailed (low-level) values to more general (high-level) concepts.
- It allows users to drill down for detailed analysis or roll up for summaries, helping simplify large datasets and improving pattern discovery.

Example of a Concept Hierarchy

hierarchy for the Location dimension.

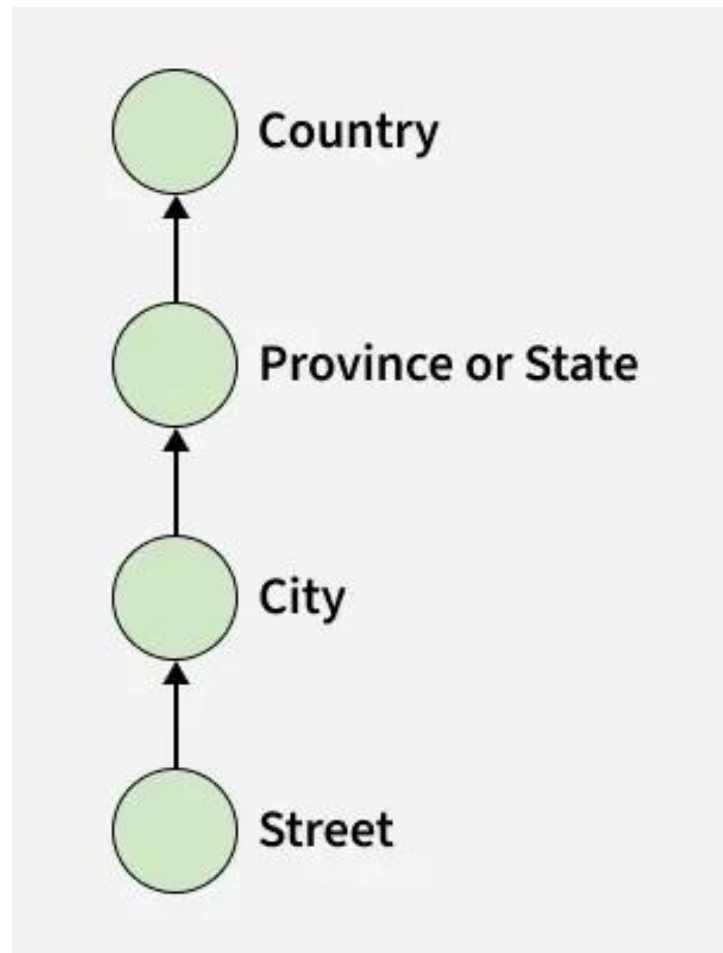
- The root node Location is generalized into Countries (USA, India).
- Countries break into States (New York, Gujarat).
- States further break down into Cities.



hierarchies allow a user to analyze data at the level they need—city-level detail or country-level summaries.

What is a Concept Hierarchy?

The hierarchical structure represents the abstraction level of the dimension location, which consists of various footprints of the dimension such as:



Example – Location Dimension



- Location → Countries → States → Cities
- Users can analyze data at city-level (detailed) or
- country-level (summary)

Applications of Concept Hierarchy

- **Data Warehousing:**
 - Organizes data into multiple levels(e.g., city → state → country) to improve reporting and OLAP analysis
- **Business Intelligence:**
 - Customer and sales analysis and Market Data at different abstraction levels for better decision making.
- **Online Retail:**
 - Product categorization and recommendations, making search, navigation, and recommendations easier.
- **Healthcare:**
 - Grouping patients by diagnosis or age to identify patterns and support clinical analysis.
- **NLP:**
 - Builds Topic hierarchies for text analysis for unstructured data.
- **Fraud Detection:**
 - Groups financial transactions into meaningful levels to Detect unusual transaction patterns.

Types of Concept Hierarchies

- **Schema-Based:**
 - Derived from Database schema (City → State → Country)
- **Set-Grouping:**
 - Values grouped by categories (Product → Sub-category → Item)
- **Operation-Derived:**
 - Based on aggregation, normalization, or range partitioning
- **Rule-Based:**
 - User-defined rules (IF income > 1M → High income; ELSE → Low/Medium income)
- **Explicit:**
 - Manually designed by domain experts of database designers (Department → Faculty → University).
- **Implicit:**
 - Automatically generated from data (Attribute values
 - Numerical ranges, Data types, Relationships in the schema)

Methods to Generate Concept Hierarchies

- Schema-Based: Uses database relationships
- Rule-Based: User-defined rules or metadata
- Data-Based: Clustering, binning, range partitioning, frequency grouping

Methods to Generate Concept Hierarchies

- **Schema-Based Generation:**

Hierarchy is derived from the database schema itself. Examples:

- Uses primary key → foreign key relationships or existing dimension tables in data warehouses.
- Automatically reflects **real-world data relationships**.
- **Example:**
 - Location dimension: City → State → Country
 - Date dimension: Day → Month → Year
- **Best for:** Structured data where schema already defines natural hierarchies

Methods to Generate Concept Hierarchies

- Rule-Based Generation:

Hierarchies are defined using **user-defined rules or metadata**.

- Analysts create **conditions or thresholds** to categorize data.

Example rule:

if age < 18 → Young; if 18–60 → Adult; else → Senior

- Useful when domain knowledge can guide grouping.

- **Example:**

- Numerical (Salary):

- IF salary $\leq$ 30k → Low
- IF salary 31–60k → Medium
- IF salary > 60k → High

- Categorical (Product):

- IF product type = 'iPhone' → Smartphones → Mobile Devices → Electronics

- **Best for:** When expert knowledge or business rules are available

Methods to Generate Concept Hierarchies

- **Data-Based Generation:**
 - Hierarchies created using data distribution. Methods include: Clustering, binning, range partitioning, frequency grouping
 - Hierarchies are generated automatically based on **data characteristics or distribution**.
 - **Clustering:** Group similar values (e.g., customers with similar spending patterns).
 - **Binning/Discretization:** Convert numeric data into intervals.
 - **Range Partitioning:** Divide numeric attributes into ranges.
 - **Frequency-based grouping:** Group categories based on occurrence frequency.
 - **Example:**
 - Age: 0–12, 13–19, 20–60, >60 → Child, Teen, Adult, Senior
 - Product sales: Frequent products → Top-selling category, Others → General category
 - **Best for:** Large datasets where patterns or natural groupings are unknown.

Why Concept Hierarchies Are Important



- **Better data analysis and pattern discovery**
 - Hierarchies organize data into manageable levels, making patterns easier to identify. They support summarization and help analysts look at data from different perspectives.
- **Improved Data exploration & visualization (drill-down / roll-up)**
 - Tree-like structures make dashboards more interactive-users can move between summary and detailed views easily through drill-down or roll-up operations.
- **Faster and more accurate algorithms**
 - Many data mining algorithms work better on generalized data. Hierarchies reduce noise and help algorithms detect clearer patterns.
- **Data cleaning & preprocessing**
 - Concept hierarchies help identify: Outliers, Inconsistent values and Missing ranges, which leads to cleaner, higher-quality datasets.
- **Better Representation domain knowledge**
 - Hierarchies capture real-world relationships (e.g., Product → Brand → Category), making the data model easier to understand.

Numerical Data – Discretization + Hierarchy

- Salary Data: 12k, 28k, 55k, 90k
- Step 1: Discretization → 0–30k, 31–60k, 61–100k
- Step 2: Concept Hierarchy → Low, Medium, High
- Reduces complexity while keeping meaningful categories

Categorical Data – Concept Hierarchy



- Product Dimension:
 - Item → Sub-category → Category → Department
- Example:
 - iPhone → Smartphones → Mobile Devices → Electronics
- Allows analysis at any abstraction level

Key Takeaways



- Organizes data from detailed → general
- Essential for data mining, OLAP, and pattern discovery
- Can be explicit, implicit, or rule-based
- Works for numerical and categorical data
- Improves visualization, analysis, algorithm performance, and domain knowledge representation